Life Insurance Mortality Risk and Premium Pricing

Second-year actuarial science portfolio project | Data: Statistics Canada life table, Canada, 2022/2024

Executive summary

Research question. How do mortality rates vary by age and sex, and how can those differences be used to estimate the cost of a term life insurance policy?

Approach. I cleaned the Statistics Canada complete life table, computed survival probabilities from the death probabilities (q_x), and priced end-of-year term life policies as the expected present value (EPV) of the death benefit, then tested how premiums respond to issue age, term, benefit and interest rate.

Headline results. For a $250,000 10-year policy at 3% interest, the annual net premium is about $444 for a male aged 35 and $234 for a female aged 35, rising to $1,719 and $1,055 at age 55. Age and term length move the premium far more than the interest rate does. These are net premiums from population mortality, not insurance quotes.

Data source

Statistics Canada Table 13-10-0114-01 (complete life table, three-year estimates; Canada and nine provinces, excluding PEI and the territories). The analysis uses **Canada, 2022/2024**, ages 0 to 110, males and females. This is a **period** table: a snapshot of mortality conditions in those years, not a cohort table and not a forecast. The cleaned q_x series has 143,190 rows across all periods, geographies and sexes, with no missing values, no duplicates, and every value within [0, 1].

Major mortality patterns

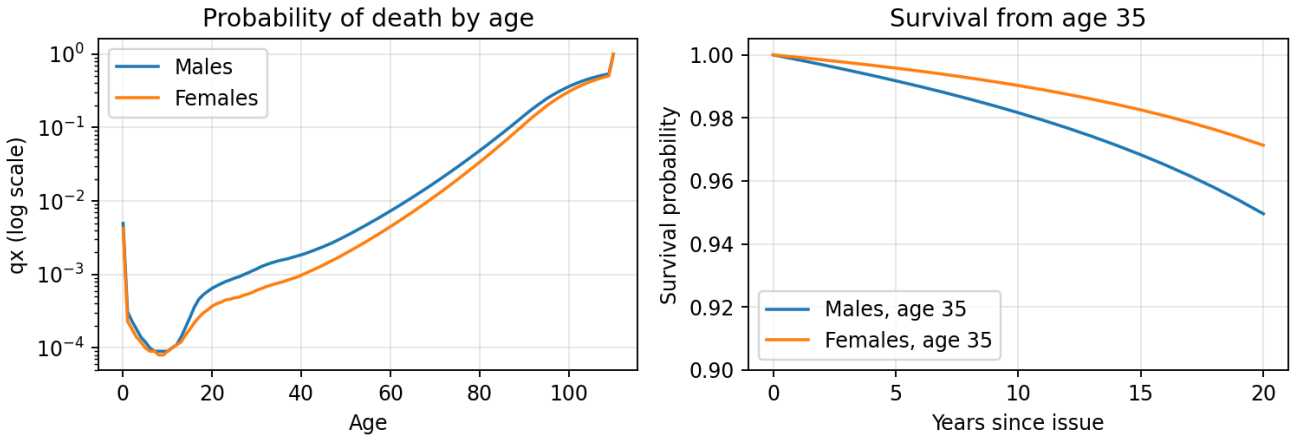


Figure 1. Left: q_x by age on a log scale. Right: survival probability over 20 years for a life aged 35.

- Mortality is low through the young-adult ages and then rises steadily; on the log scale the rise is close to a straight line over the middle ages, which is consistent with roughly exponential growth in q_x.
- At every age shown, the male q_x is higher than the female q_x. The male-to-female ratio is about 1.85 at age 25, 2.0 at 35, 1.8 at 45 and 1.7 at 55, narrowing at older ages (about 1.5 at 75). This describes the table; it does not explain why.

Probability of surviving 10 and 20 years from the issue age (Canada, 2022/2024):

Issue age	Male 10-yr	Male 20-yr	Female 10-yr	Female 20-yr
25	98.8%	97.0%	99.4%	98.4%
35	98.2%	95.0%	99.0%	97.1%
45	96.7%	89.9%	98.1%	93.8%
55	93.0%	77.7%	95.6%	85.1%

Data preparation and actuarial method

Cleaning (src/clean_data.py). The raw file is left unchanged. I kept the Death probability (q_x) element, extracted the numeric age from the age-group label, and checked for missing values, duplicates and values outside [0, 1]; none were found. The other life-table elements (l_x, d_x, e_x, margins of error) are not used.

Model. For an n-year term policy with benefit B, annual effective interest rate i and issue age x, with ${}_tp_x$ the probability of surviving t years:

EPV of death benefit (= net single premium) = sum for t = 1..n of $B\,(1+i)^{-t}\,{}_{t-1}p_x\,q_{x+t-1}$

Temporary annuity-due = sum for t = 0..n-1 of $(1+i)^{-t}\,{}_tp_x$

Annual net premium = EPV divided by the annuity-due

Death benefits are paid at the end of the year of death; premiums are paid at the start of each year while the insured is alive. The net premium excludes expenses, profit, lapses, taxes, underwriting and capital.

Premium comparison

Annual net premium, $250,000 benefit, 3% interest, Canada 2022/2024:

Sex	Issue age	10-year	15-year	20-year	10-yr net single
Male	25	$280	$314	$349	$2,447
Male	35	$444	$505	$590	$3,875
Male	45	$790	$958	$1,176	$6,854
Male	55	$1,719	$2,145	$2,696	$14,717
Female	25	$143	$161	$182	$1,256
Female	35	$234	$275	$330	$2,049
Female	45	$460	$567	$710	$4,011
Female	55	$1,055	$1,345	$1,737	$9,126

Sensitivity analysis

I varied issue age (20 to 65), term (10, 15, 20 years), benefit ($100,000 to $1,000,000) and interest rate (1% to 7%) for both sexes, giving 7,728 combinations.

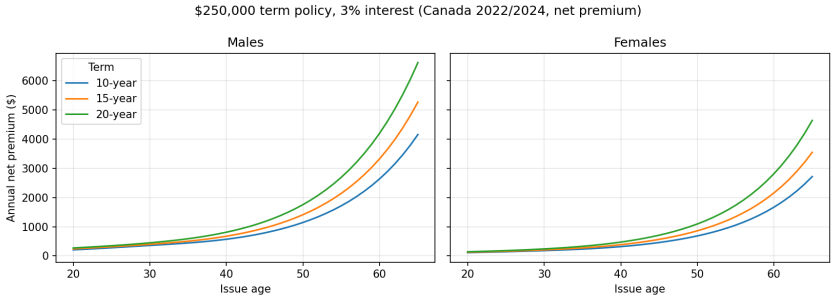


Figure 2. Annual net premium by issue age and term, $250,000 benefit, 3% interest.

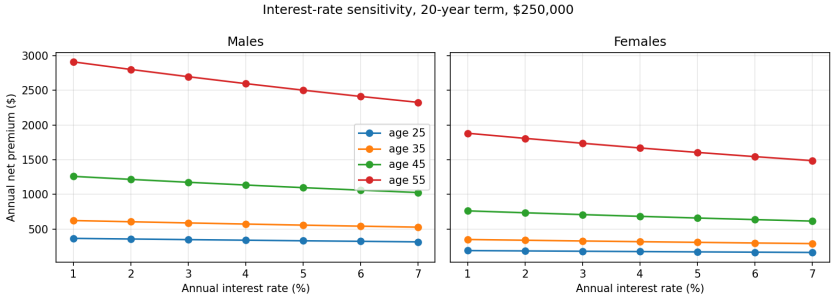


Figure 3. Annual net premium by interest rate for a 20-year, $250,000 policy.

- **Age** is the dominant driver: the male 10-year premium is about six times higher at 55 than at 25.
- **Term and benefit:** longer terms cost more per year because they cover older ages; premiums scale in direct proportion to the benefit.
- **Interest rate:** raising the rate from 1% to 7% lowers the annual premium by about 18.5% for a male aged 45 on a 20-year term ($1,260 to $1,027), a smaller effect than age.

Validation

- **Unit and consistency tests (7, all passing):** survival starts at 1, never increases and stays in [0, 1]; zero mortality gives zero EPV; premiums scale proportionally with the benefit; certain death in year one gives a premium of 1/(1+i); the term identity $A + v^n {}_np_x = 1 - d$ (annuity-due) holds; the level premium times the annuity equals the net single premium; higher interest lowers the premium. The tests use a hypothetical mortality curve, so they do not depend on the data file.
- **Independent Excel recalculation:** two cases were rebuilt row by row with spreadsheet formulas and match the Python results to the cent: male, age 35, 10-year ($3,875.27 net single, $444.27 annual) and female, age 45, 20-year ($10,672.07 and $709.73). Both use the same q_x source, so this checks the formulas, not the data.

Limitations

- **Population is not insured lives.** Insured-life mortality differs because of underwriting, selection and socioeconomic factors. Real premiums are not expected to match these figures.
- **No mortality improvement.** A period table is a snapshot; a 20-year policy priced on it ignores future changes in mortality.
- **Old-age estimates are volatile** because data are sparse at the oldest ages. The policies priced here mostly avoid those ages.
- **Overlapping periods.** Adjacent three-year windows share two years of data and are not independent. The 2020/2022 to 2021/2023 windows include COVID-19 years; the choice of 2022/2024 is a judgment, not a neutral default.
- **Interest rates.** A single constant rate is a simplification; actual rates change over a policy's life.
- **Net premium only:** no expenses, profit, lapses, taxes, underwriting adjustments or capital requirements.
- **Coverage:** the source table excludes PEI and the territories.
- **Sensitivity is not accuracy.** The premium comparisons show how the model responds to its inputs; they are not evidence that the model predicts real insurance costs.

Implications

Within this simple deterministic model, the differences in mortality by age and sex translate into large differences in the cost of term cover, and the gap between the sexes in the table carries through to the premiums. The model is a reasonable teaching illustration of how survival probabilities, discounting and expected present value combine into a premium. It is not a basis for pricing real policies.

Possible improvements

- **Out-of-sample test:** project mortality from an earlier table and compare against observed later rates.
- **Mortality improvement:** add an explicit improvement assumption instead of a fixed period table.
- **Expense loading and lapses** to move from net to gross premiums; compare provinces; test an insured-life table if one is available.

Reproducibility

Code, tests, the Excel check, the sensitivity grid and an interactive Streamlit dashboard (Life Insurance Pricing Explorer) are in the project repository; the README gives the commands to reproduce every number in this report.